

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Digital Signal Processing	Course Code:	EE3031
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	03-April 2025	Total Marks:	20
	Section:	BEE-6A	Weight:	3
	Exam Type:	Assignment- 2 (PBL)	Page(s):	1
			CLO #	3

Student Name: _____ **Roll No.** _____ **Section:** _____

Instructions: 1. Submission instructions depending on whether it is a hardcopy/softcopy submission
2. _____

Question No.1 (CLO No. 3)

Marks: 10

- a) Consider the following periodic sequence

$$x_p(n) = \cos \frac{2\pi}{10} n \quad -\infty < n < \infty$$

with frequency $f_o = \frac{1}{10}$ and fundamental period $N = 10$. **Compute** the 10-point DFT of the sequence $x(n) = x_p(n), 0 \leq n \leq N - 1$.

- b) Why DFT is able to extract spectral information from $x(n)$?

Question No. 2 (CLO No. 3)

Marks: 10

Let $x(n)$ be an N-point sequence with an N-point DFT $X(k)$ (N even)

- a) Consider the time-aliased sequence

$$y(n) = \begin{cases} \sum_{l=-\infty}^{\infty} x(n + lM), & 0 \leq n \leq N - 1 \\ 0 & \text{elsewhere} \end{cases}$$

Determine the relationship between the M-point DFT $Y(k)$ of $y(n)$ and the Fourier transform $X(w)$ of $x(n)$.

- b) Why N-point DFT is able to extract information from time-aliased $\sum_{l=-\infty}^{\infty} x(n + lM)$.